Issue	**1**
Factory ID	**26207**
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Arshad Corporation (Pvt.) Ltd. (Apparel Division)

4.5 KM Jaranwala Road, Khurrianwala
Faisalabad
Punjab Pakistan

Latitude: [31.471655], longitude: [73.309928]
Date of inspection: 21-Apr-2026

Electrical Safety Inspection Report

Observations & Actions



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Pakistan Accord Initial Safety Inspection Report

About this report:

This report has been prepared by the Health and Safety Accord Pvt Ltd after visiting the factory on 21-04-2026 as part of the inspection and remediation program of the Pakistan Accord on Health and Safety in the Textile and Garment Industry. The purpose of the report is to provide key findings and identify related safety improvements needed at the inspected factories. This inspection report may not identify minor deficiencies, which may be further addressed as part of follow-up inspections.

In addition, while this round of initial inspections covers fire, electrical and structural safety, the Pakistan Accord intends to include inspections on boiler safety and hazardous material in later rounds.

This report will serve to facilitate discussions between the Pakistan Accord, the factories and the responsible brands that source from this facility about the remediation requirements. An implementation schedule shall be developed by the factory in the form of a Corrective Action Plan (CAP) to remediate each of the findings identified in the inspection report, including timelines and cost estimates, for approval by the Chief Safety Officer (CSO).

Applied Standards:

This safety assessment was conducted by the Pakistan Accord with reference to the Pakistan Accord Building Standard.

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1. Executive Summary

On 21-Apr-2026, Umer Sajjad and Sarah Aiman on behalf of the Pakistan Accord carried out an initial electrical inspection for Arshad Corporation (Pvt.) Ltd. (Apparel Division), a factory at the address and coordinates given on the cover page of this report. The inspection was carried out in accordance with the Pakistan Accord Building standards.

We met with Mr. Saleh Muhammad (Manager), Mr. Rizwan Akram (Deputy Manager – Electrical), Mr. Zeeshan Sajid (Compliance Officer), Mr. Muhammad Salman Sarwar (Civil Engineer) and other staff members of Arshad Corporation Pvt. Ltd. (Apparel Division). The assessment was conducted as a part of an overall Structural, Fire, and Electrical initial inspection.
The factory operates in single general shift. Factory employs approximately 500 workers with approximately 6% female representation.
The factory comprises of two main buildings and five ancillary structures owned by Arshad Corporation (Pvt.) Ltd (Apparel Division). During the inspection, the building highlighted in blue in Figure 1.1 were surveyed with the site boundary indicated in red.

- **Main Building** is a two-storey (Ground + 1) building. The building is used for stitching, workshops, sample room and offices (First Floor), and offices, fabric store, cutting, accessory store, finishing and packing hall, warehouse, rest areas (Ground Floor).
- **Canteen Building** is a single-storey (Ground Only) building. The building finishing works are ongoing, and its future intended use is dining area and kitchen (Ground Floor).
- **Ancillary Structures** consist of the following:
  - **Security Office** is a single storey (Ground Only) structure. The structure is used for offices (Ground Floor).
  - **Overhead Water Tank (OHWT)** is an elevated structure. The structure is used for water storage.
  - **Office Block** is a single storey (Ground Only) structure. The structure is used for offices, childcare room, female security room (Ground Floor).
  - **Toilet Block 1** is a single storey (Ground Only) structure. The structure is used for ladies / gents toilet (Ground Floor).
  - **Toilet Block 2** is a single storey (Ground Only) structure. The structure is used for gents toilet (Ground Floor).

A summary of the main concerns in relation of electrical safety systems identified during the inspection (non-exhaustive) that should be actioned are as follows:

• **Abnormal Temperature:** Within electrical panels abnormal temperatures were recorded at various locations (refer section 7.2 for further details). Hotspots within panels can cause short circuits and fire due to insulation damage and overheating of cables and electrical components.
• **Fire Alarm Detection System:** An addressable fire alarm system was installed in the facility in the security room that serves the apparel division of the facility. During the inspection, several deficiencies were noted, including the use of non-listed panel and components, inadequate detection coverage, missing manual call points (MCP's), fault indication on panel, faulty manual call points (MCP's), 220V manual hooters, non-fire rated cabling, incorrect placement of manual call points (MCP's) and detection devices, low audibility in several areas during testing, all of which pose a significant life-safety risk during a fire emergency.
• **Earth Missing:** In most areas, earth connection for distribution boards, electrical machines and grounding/bonding of electrical panels, electrical

machines, cable trays, metal structures, metallic panel doors and metal furniture & fixings, were missing. The earthing system is important for workers operating machines, in case of short circuit/fault condition.

• **Emergency lighting System:** The emergency lighting system had inadequate coverage and was non-functional in most areas of the facility hindering safe evacuation during power outages and increasing the risk of accidents.

We see no reason to suspend operations in the facility due to these concerns, subject to the actions identified being addressed within the specified timeframes.

Our Limitations and Assumptions are also noted at the end of this report.

For detailed analysis of the above, please refer to the full inspection report below, which further details each issue, and also includes photographs. Further actions with associated priorities and timeframes are also detailed with each issue. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

2. Building Extents



Figure 1.1. Satellite view of the factory – Red dashed line indicates site boundary, blue highlight indicates area inspected

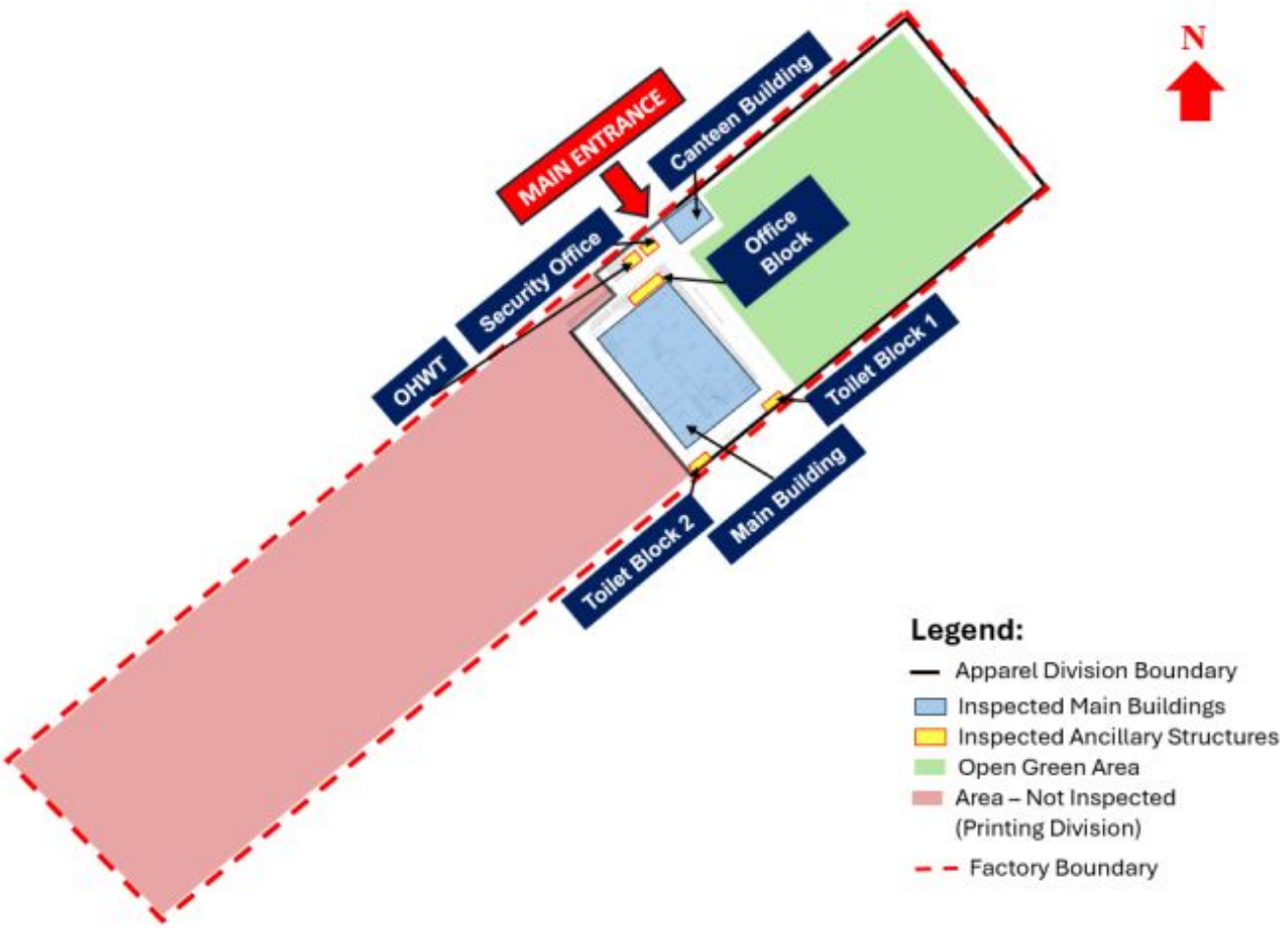


Figure 1.2. Site layout - Red dashed line indicates site boundary

2.1 Building Use

The site layout consists of the following two buildings and five ancillary structures:

Building 1 (Main Building):
L00 – Ground floor: Offices, fabric store, cutting, accessory store, finishing and packing hall, warehouse and rest areas.
L01 – First floor: Stitching, workshops, sample room and offices.
Building 2 (Canteen Building):
L00 – Ground floor: Dining area and kitchen (finishing works ongoing).

Ancillary Structures:
Security Office:
L00 – Ground floor: Offices.
Overhead Water Tank - OHWT:
L00 – Ground level: Water storage.
Office Block:
L00 – Ground floor: Offices, childcare room, female security room.
Toilet Block 1:
L00 – Ground floor: Ladies and gents toilets.
Toilet Block 2:
L00 – Ground floor: Gents toilets.

2.2 Utility Source

The facility operates with 400 V low voltage supply received from the neighboring printing division of the Arshad Corporation (Pvt.) Ltd. This supply is received via underground low voltage cables at main distribution board located at ground floor of the main building. The electrical supply from this main distribution board is then distributed throughout the facility via power cables.

2.3 Backup source

The facility backup supply depends on the neighboring printing division of the Arshad Corporation (Pvt.) Ltd. The details of the backup supply are not provided. Both the emergency lighting system and fire alarm and detection system (FADS) have internal battery backups available.

3. Documentation

This table shows the detail of the electrical documentation provided by the factory.

S.NO	Description	Availabe (Y/N)	Created Date	Status / Remarks
1	As-Built Layout Drawing	No		
2	Single Line Diagrams	Yes	Feb 2018	SLD were not up-to-date and dosn't comply with relevent standards.
3	Fire Alarm Drawing	No		
4	Earth Test Report	No		
5	Thermographic Test Report	No		
6	Insulation Resistance Test Report	No		
7	LPS Risk Assessments	No		
8	Periodic Maintenance Record	No		
9	Transformer Oil Testing Report	No		

4. Electrical Safety Measures

During our electrical site inspection of Arshad Corporation (Pvt.) Ltd. (Apparel Division), a number of electrical issues were identified.

It is essential to address these electrical safety issues promptly to enhance safety and compliance within Arshad Corporation (Pvt.) Ltd. (Apparel Division). Each issue should be carefully evaluated, and the recommendations provided in this report should be followed diligently to rectify these concerns. The safety and well-being of the factory's employees depend on these corrective actions. Where issues are identified in the report, they are grouped by topic, not in order of importance. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

Recommendations in this report are prioritized as follows:

Immediate: These recommendations address safety and compliance concerns that require immediate attention.

Medium term (within approx. 6-weeks): Recommendations of high importance but needing slightly more time to implement.

Longer term (within approx. 6-months): Important recommendations that require a longer period to fully implement or that enhance long-term safety and efficiency.

All survey, testing, and design activities specified in this report shall be performed by a qualified electrical engineer. The recommended installation and modification tasks should be executed by a licensed professional electrician.

Before commencing any installation and/or modification works, the proposed design must be submitted to the Pakistan Accord for approval.

Upon completion of the works, the necessary evidence required to verify that the tasks have been completed to the agreed standards will be determined during the development of the Corrective Action Plan (CAP).

5. Remediation Support

5.1 Engineering Support

Some items in the Corrective Action Plan (CAP) will require remediation by a qualified engineer. Please refer to the Remediation Plan for guidance on which tasks can be completed using internal resources and which require engineering support. F-202604-066158

Timeline	Remediation Plan
Medium Term CAP-0079000	Factories should begin remediation of CAP items – such as cleaning, cable organization, and basic corrections – using internal resources. For systems or design-related remediation, engagement of a qualified Electrical Engineer is required.

6. Diagrams & Drawings

6.1 Sub-station room layout drg

This section refers to the electrical rooms within the facility.

During the inspection conducted, it was observed that there were no substation room layout drawings available for the factory.

There are no visible indications of substation equipment locations or pathways for maintenance and emergency access.

The absence of substation room layout drawings poses significant risks to life and fire safety within the factory premises. Without these critical documents, emergency responders and maintenance personnel may face difficulties in locating and isolating electrical faults during emergencies. This delay can result in prolonged exposure to electrical hazards, increasing the risk of electrical fires and endangering lives. Furthermore, the lack of layout drawings complicates routine maintenance, potentially leading to undetected faults and increased likelihood of equipment failure. F-202604-066052

Timeline	Remediation Plan
Medium Term CAP-0078808	Develop detailed substation room layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). Ensure that all critical equipment and pathways are accurately documented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0078809	Regularly update and review the substation room layout drawings to reflect any changes in equipment or configuration. Provide training sessions for maintenance personnel and emergency responders on how to interpret and utilize the layout drawings.

6.2 Single Line Diagram (SLD)

This section refers to the entire facility.

It was found that the electrical single line diagrams (SLDs) for the factory were missing.

This deficiency was noted in all electrical rooms and areas where SLDs should be prominently displayed.

The factory lacks documented diagrams that illustrate the electrical distribution system, including the connection and configuration of all electrical components.

Single line diagrams are crucial for understanding the electrical distribution network and for troubleshooting issues effectively. Without these diagrams, maintenance personnel and emergency responders lack critical information necessary for isolating faults and managing power during emergencies. This can lead to delays in addressing electrical issues, increasing the risk of electrical fires and potential harm to personnel. Moreover, the lack of SLDs complicates routine maintenance and inspection processes, potentially resulting in undetected electrical faults and system inefficiencies.F-202604-066055

Timeline	Remediation Plan
Medium Term CAP-0078812	Develop comprehensive electrical single line diagrams for the entire facility in accordance with the requirements outlined in Section 514.9.1 of BS7671 18th Edition + A2 (2022). Ensure all components and connections are accurately represented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0078813	Implement a regular review and update process for the single line diagrams to reflect any changes in the electrical system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the diagrams.

6.3 Electrical layout drawing

This section refers to the entire facility.

During the survey, it was noted that the factory lacked electrical layout drawings - including small power layouts, lighting layouts, and containment layouts.

This absence was consistent across all areas where such documentation should be available.

These drawings are vital for understanding the placement and connections of electrical systems, which is crucial for identifying and mitigating potential hazards. In emergencies, the lack of detailed layouts can delay the identification and isolation of electrical faults, increasing the risk of fire and electrical accidents. Additionally, without proper layout drawings, routine inspections and maintenance may overlook critical issues, potentially leading to system failures and safety breaches.F-202604-066062

Timeline	Remediation Plan
Medium Term CAP-0078817	Develop comprehensive electrical layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). To avoid drawings becoming overcrowded, separate drawings shall be prepared for each electrical system. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0078818	Implement a regular review and update process for the electrical layout drawings to reflect any changes in the system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the layouts.

7. DBs & Elec Protection Systems

7.1 Circuit breakers

This section refers to the entire facility.

During the survey the circuit breakers installed throughout the facility inside the electrical panels and distribution boards were inspected.

i. It was found that a single-phase load was fed from a three-pole breaker. Using a single phase of a three-pole breaker can lead to unbalanced load conditions. This can cause the breaker to trip unexpectedly or, in some cases, fail to trip when necessary, compromising the safety of the system. This issue was specifically observed at main distribution board ground floor main building.

ii. It was also found that the circuit breaker was installed in outdoor areas without weather-protective enclosures, posing a risk of short circuits or electrical incidents during adverse weather conditions. This issue was observed outside main building.

iii. It was observed that multiple circuit breakers had ratings that were not appropriate for the connected cable sizes such as:
a) 95 mm² cable connected with 400 A circuit breaker at main distribution board ground floor main building.
b) 35 mm² cable connected with 200 A circuit breaker at main distribution board first floor main building.
c) 25 mm² cable connected with 150 A circuit breaker at panel outside main building.
d) 16 mm² cable connected with 100 A circuit breaker at compressor room main building.

Specifically, some circuit breakers had higher ratings than the cables could safely handle, which poses a significant risk of overheating and potential fire hazards. Conversely, other circuit breakers had lower ratings than necessary, resulting in frequent and unnecessary tripping. To ensure both safety and operational efficiency, corrective action is required to align the circuit breaker ratings with the cable capacities. This adjustment will mitigate the risks associated with overheating and improve the reliability of the electrical system. F-202604-066068

Timeline	Remediation Plan
Medium Term CAP-0078654	i. Remove single-phase connections from three-pole breakers and replace them with appropriate single-pole breakers, ensuring compliance with standard current-carrying capacities and preventing load imbalance.

Medium Term CAP-0078655	ii. Install the circuit breakers inside a properly rated IP weatherproof enclosure with secure mechanical mounting and ensure all terminations are adequately sealed and protected against environmental exposure.
Medium Term CAP-0078656	iii. Evaluate the breaker ratings and cable sizes for all circuits in the facility according to Section 433.1.1 and Appendix 4 of BS 7671 18th Edition + A2 (2022). Ensure that the breaker rating does not exceed the cable's current-carrying capacity. If the breaker has adjustable settings, they must be configured to allow current within the cable's ampacity after applying derating factors.



Figure 2.1 - Single phase loads fed from a three pole circuit breaker at main distribution board ground floor main building



Figure 2.3 - 35 mm² cable with 200 A circuit breaker at first floor main building



Figure 2.2 - Circuit breaker installed outdoor without weatherproof enclosure outside main building



Figure 2.4 - 95 mm² cable with 400 A circuit breaker at main distribution board ground floor main building

7.2 Hot Spots

This relates to the entire facility.

During the inspection, Thermographic surveys were conducted on the electrical boards.

The survey identified numerous hot spots within panels and distribution boards. Hot spots in electrical distribution systems can be caused by several factors, including:

> Loose connections
> Overloaded circuits
> Faulty components
> Improper sized cable
> Insulation degradation

It was also found that the thermographic report was not available at the site. Without thermography testing, faults such as overheating components or loose connections may go unnoticed, which increases the risk of fire. Early detection through thermography can prevent fire incidents.

The categories of identified hot spots have been evaluated as per the ANSI NETA MTS 2019 standard, Table 100.18. Hot spots were observed, particularly at the following locations:
- 101°C on circuit breaker at main distribution board ground floor main building.
- 101°C and 96.1°C on circuit breaker termination points at main distribution board first floor main building.
- 82.4°C on cable in distribution board at finishing area ground floor main building.
- 82°C on circuit breaker in distribution board at cutting area ground floor main building.
- 73.9°C on circuit breaker termination point in distribution board at compressor room ground floor main building.

Circuit breakers are typically designed to operate safely within an ambient temperature range of -20°C to 40°C. Commonly used PVC insulated cables typically have a maximum operating temperature of 70°C. Prolonged exposure to temperatures above these can degrade the insulation of the breaker as well as the cable and result in reduced performance.

Hot spots in electrical systems pose significant life safety risks due to several factors. Overheated components can ignite surrounding materials, leading to electrical fires. This is particularly dangerous in densely packed panels and distribution boards where flammable materials may be present. Hot spots can degrade insulation on wires and components, increasing the risk of electrical shock to personnel who might come into

contact with exposed conductive parts. Overheating can contribute to conditions that might trigger an arc flash, a dangerous high-temperature event that can cause severe injuries or fatalities to personnel working near the affected equipment.F-202604-066051

Timeline	Remediation Plan
Immediate CAP-0078673	Any circuits identified on site with temperatures exceeding 40°C above the ambient temperature, operating above 90°C or exhibiting visual signs of burning must be immediately isolated and taken out of service. These circuits should remain de-energized until a thorough review has been conducted and appropriate remedial actions have been implemented to resolve the underlying issues.
Medium Term CAP-0078674	During the inspection, several circuits were identified where the temperature was exceeding 20°C above ambient temperature. It is recommended that the factory conducts comprehensive thermographic surveys of all electrical panels to detect similar issues. Identified hot spots should be thoroughly reviewed to determine their underlying causes. Corrective actions must be taken to address and resolve these issues, ensuring safe and reliable operation of the electrical systems.
Longer Term CAP-0078675	Regular thermographic inspections should be incorporated into the maintenance schedule to proactively manage and mitigate the risks associated with overheating.

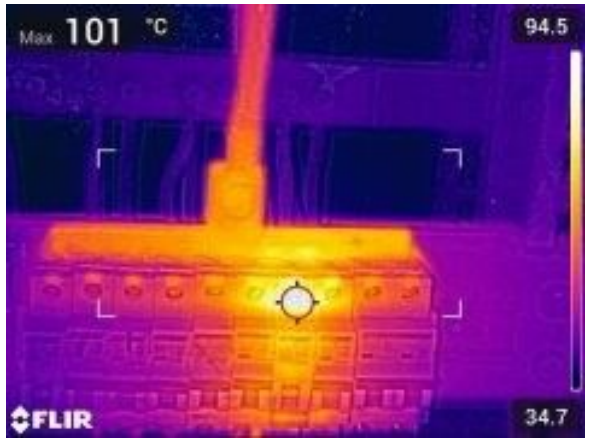


Figure 2.5 - Hotspot found on circuit breaker in main distribution board ground floor main building



Figure 2.6 - Hotspot found on circuit breaker in main distribution board ground floor main building

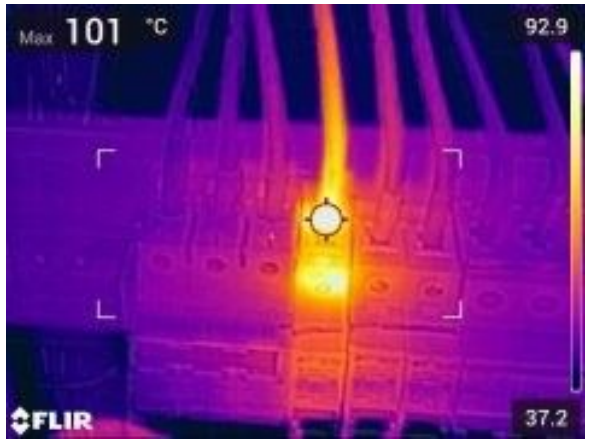


Figure 2.7 - Hotspot found on circuit breaker termination point at first floor main building



Figure 2.8 - Hotspot found on circuit breaker termination point at first floor main building



Figure 2.9 - Hotspot found on circuit breaker termination point at first floor main building



Figure 2.10 - Hotspot found on circuit breaker termination point at first floor main building



Figure 2.11 - Hotspot found on cable in distribution board at finishing area ground floor



Figure 2.12 - Hotspot found on cable in distribution board at finishing area ground floor

7.3 Earthing system

This relates to the entire facility.

During the inspection of the facility's electrical panels, distribution boards, and machinery, several observations with the earthing system were identified.

i. No earthing system design documentation or records of earth pit testing and maintenance were provided by the factory, indicating a lack of verification of the earthing system’s compliance and ongoing performance in accordance with relevant standards.
ii. The earth pits were inaccessible and unsuitably located. This issue was observed near main distribution board ground floor main building.
iii. The earthing cables were not appropriately sized according to the relevant earthing standards.
iv. The earthing system was absent in multiple locations. Several distribution boards lacked earthing cables, and many electrical machines did not have proper earth connections. This issue was observed at finishing area distribution boards ground floor, compressor room, water pumps, cutting area machines, stitching area first floor main building and canteen building.
v. Equipotential bonding for the cable trays and metallic panel covers was missing. This issue was observed at main distribution board ground floor and first floor main building.

The lack of proper earthing system throughout the facility poses substantial safety risks, such as electric shocks, fires, and equipment damage. Implementing appropriate earthing measures is crucial to prevent hazardous voltage levels and ensure operational safety. F-202604-066050

Timeline	Remediation Plan
Medium Term CAP-0078717	i. Design a complete earthing system for the facility and prepare earthing layout drawings. Submit the design and drawings to Pakistan Accord for review. Also, provide earthing system testing and maintenance records for Pakistan Accord review.
Medium Term CAP-0078718	ii. Relocate and redesign earth pits to make them easily accessible for maintenance and inspection, complying with BS 7671 18th Edition + A2 (2022) Chapter 54.
Medium Term CAP-0078719	iii. Replace all earthing cables with appropriately sized cables as per the requirements of BS 7671 18th Edition + A2 (2022) Appendix 4, Section 5. Verify that all earthing conductors meet the minimum cross-sectional area specified.

Medium Term CAP-0078720	iv. Install earthing systems where absent, ensuring compliance with BS 7671 18th Edition + A2 (2022) Chapter 54. Fit all electrical boards with the required earthing cables. Ensure that earthing conductors are properly connected and comply with BS 7671 18th Edition + A2 (2022) Section 543.1.
Medium Term CAP-0078721	v. Implement equipotential bonding for all cable trays to comply with BS 7671 Regulation 411.3.1.2. Ensure all exposed conductive parts are bonded to the main earthing terminal.
Longer Term CAP-0078722	Establish a schedule for regular maintenance and inspection of the earthing system and protective devices as per BS 7671 Regulation 651.2. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on proper earthing practices and the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.13 - Equipotential bonding missing for panel door at main distribution board ground floor main building



Figure 2.15 - No earthing system in distribution board at first floor stitching area main building



Figure 2.14 - No earth distribution available for machines at finishing area main building



Figure 2.16 - Earth pit inaccessible near main distribution board ground floor main building

7.4 Panel/cabling/busbar

This relates to the entire facility.

The visual site inspection included an examination of the electrical panels, cables and busbars throughout the facility.

i. Several electrical panels exhibited signs of corrosion, indicating potential exposure to moisture or other corrosive elements. This issue was observed at main distribution board and cutting hall ground floor main building.
ii. In the finishing hall electrical panel, it was observed that the neutral bar was not fixed properly posing short circuit and electrocution risk in case of contact with enclosure or human contact.
iii. There were visible signs of damage to panels such as broken door hinges for electrical panels particularly observed for water chiller distribution board at cutting hall ground floor main building.
iv. Phase separators were missing in some panels posing the risk of accidental contact between phases. This issue was systematic in most of the distribution boards and electrical panels throughout the facility,

The utilisation of electrical equipment that is either unsuitable for its operating environment or physically damaged poses significant safety risks. These risks include potential electrical failures, increased likelihood of short circuits, and the heightened risk of fire and electrical shock. F-202604-066065

Timeline	Remediation Plan
Medium Term CAP-0078742	i. Replace or refurbish corroded electrical panels to prevent further degradation and potential safety hazards. Implement protective measures such as installing weatherproof enclosures or using corrosion-resistant materials.
Medium Term CAP-0078743	ii. Securely fix and insulate the neutral bar inside the electrical panels throughout the facility to eliminate the risk of short circuit and accidental electrocution.
Medium Term CAP-0078744	iii. Inspect and repair or replace all damaged panels, cables, and busbars to ensure the integrity of the electrical system in accordance with Section 522 of BS7671 18th Edition + A2 (2022)
Medium Term CAP-0078745	iv. Install phase separators between adjacent live conductors of different phases within distribution boards to ensure proper segregation and protection against accidental contact, in accordance with BS 7671 Regulation 416.2.

Longer Term CAP-0078746	Establish a schedule for regular maintenance and inspection of the panel, cabling and busbars. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.17 - Phase separators missing for boiler distribution board ground floor main building



Figure 2.19 - Neutral bar not fixed properly at finishing area ground floor main building



Figure 2.18 - Broken door hinges for panel observed at cutting hall



Figure 2.20 - Corroded panel observed at ground floor main building

7.5 Loop Connection

This relates to the entire facility.

During the inspection loop connections were identified within the electrical panels.

In these loop connections, one circuit breaker is energised by the live part of another circuit breaker, creating loops within the panel instead of using a busbar system.

This issue was observed at cutting hall ground floor, stitching area first floor and finishing area of main building, canteen building and security room.

Avoiding loop connections is crucial due to several significant risks. Loop connections can complicate maintenance activities, making it difficult to trace and isolate circuits. These connections can lead to increased resistance and overheating, potentially damaging components. Loop connections raise the risk of accidental phase mixing, which can result in short circuits. The above issues compromise personal safety and significantly increase the risk of fire.F-202604-066058

Timeline	Remediation Plan
Medium Term CAP-0078752	All electrical panels shall be modified to eliminate loop connections. A busbar system within the electrical panels shall be utilised to provide each circuit with the required live, neutral, and earth connections.



Figure 2.21 - Loop connection inside distribution board at canteen building

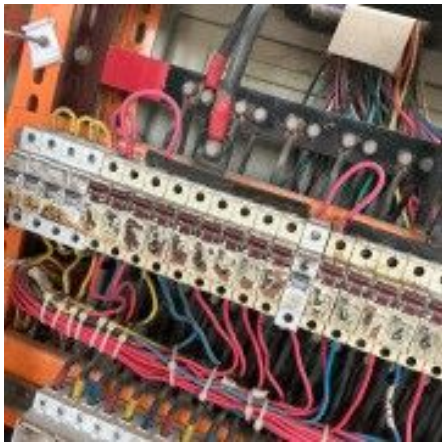


Figure 2.22 - Loop connection inside distribution board at stitching hall first floor main building



Figure 2.23 - Loop connection inside distribution board at cutting hall main building



Figure 2.24 - Loop connection inside distribution board at security room

7.6 Multiple Connections

This relates to the whole facility.

During our inspection, multilooping was observed in a number of electrical panels across the facility.

In these cases, multiple circuits were being protected by a single circuit breaker.

This issue was observed at main distribution board ground floor, boiler room, cutting hall, finishing and stitching area of main building, canteen building and fire pump area.

Multilooping can result in improper distribution of electrical current, leading to voltage drops and potential electrical hazards. To mitigate these risks and ensure the safety and reliability of the electrical system, it is essential to adhere to proper installation practices. This includes using appropriate dedicated circuit breakers for each circuit. F-202604-066057

Timeline	Remediation Plan
Medium Term CAP-0078757	The electrical installation throughout the facility shall be modified to ensure each circuit is protected by a dedicated circuit breaker of a suitable type and rating for each individual circuit in accordance with Section 430 of BS 7671 18th Edition + A2 (2022)



Figure 2.25 - Multiple connections in distribution board at cutting hall main building



Figure 2.26 - Multiple connections in distribution board at stitching hall main building



Figure 2.27 - Multiple connections in distribution board at finishing area main building



Figure 2.28 - Multiple connections in distribution board at fire pump area

7.7 Openings

This relates to the entire facility of the factory.

During the visual inspection, the electrical panels and distribution boards were examined for unsealed openings to ensure compliance with the IP rating requirements.

Unsealed openings were discovered in the panels and distribution boards. Additionally, the base plate on several electrical panels was missing.

This issue was observed at main distribution board ground floor, compressor room, cutting hall, fire pump distribution board, stitching area first floor main building and outside main building.

These deficiencies pose multiple hazards. Unsealed openings and missing panels expose live electrical components, significantly increasing the risk of accidental contact and electric shock to personnel working nearby. The unsealed openings allow dust, debris, and moisture to enter, which can lead to short circuits and increase the risk of fire. The absence of proper sealing compromises the IP rating of the panels, which is crucial for ensuring safety and protection against environmental factors.F-202604-066049

Timeline	Remediation Plan
Medium Term CAP-0078761	Corrective actions are required to seal all openings and replace missing panels to restore the integrity and safety of the electrical systems. The electrical system shall ensure compliance with Sections 614.2.1, 412.2.2, and 512.2 of BS 7671 18th Edition + A2 (2022).
Longer Term CAP-0078762	Conduct regular maintenance and maintain comprehensive records for all electrical panels to ensure there are no openings and that the IP ratings of the panels remain uncompromised.



Figure 2.29 - Unsealed openings inside main distribution board ground floor main building



Figure 2.30 - Unsealed openings inside distribution board at stitching area main building



Figure 2.31 - Unsealed openings inside distribution board outside main building



Figure 2.32 - Unsealed openings inside distribution board at fire pump area

7.8 Lint & dust inside panel board

This relates to the entire facility.

During the inspection, the cleanliness of electrical panels, distribution boards and electrical containment was assessed.

The inspection revealed a significant accumulation of lint and dust in most of the distribution boards, panels and trunking systems in main distribution board ground floor, first floor main building and security room.

The presence of lint and dust inside distribution boards poses several hazards. Accumulated lint and dust can act as fuel for fires if exposed to electrical currents or sparks, significantly increasing the risk of electrical fires. The cooling capacity of electrical components can also be impaired by the presence of dust and lint, leading to overheating.

Additionally, the accumulation of dust and lint can obstruct proper operation and maintenance, thereby impairing the functionality of the electrical components.F-202604-066063

Timeline	Remediation Plan
Medium Term CAP-0078763	All electrical panels and trunking runs shall be inspected and cleaned of any dust or lint in accordance with Section 522.4.2 of BS7671 18th Edition + A2 (2022)
Longer Term CAP-0078764	A regular maintenance procedure shall be established to ensure routine inspection and cleaning of all electrical panels and trunking runs. This will prevent the accumulation of lint and dust, thereby maintaining the safety and efficiency of the electrical systems.



Figure 2.33 - Lint and dust inside main distribution board ground floor main building



Figure 2.34 - Lint and dust inside distribution board at first floor main building



Figure 2.35 - Lint and dust inside distribution board at security room

7.9 Lightning Protection System (LPS)

During the inspection, the lightning protection strategy of the factory was assessed.

No lightning protection system was installed, and there was no risk assessment provided to confirm whether it was required or not.

Lightning strikes can cause significant damage to buildings, primarily through the ignition of fires. When lightning strikes a structure, it can cause an electrical surge that ignites flammable materials. Without a proper lightning protection system (LPS), the energy from a lightning strike can travel through the building's electrical and plumbing systems, creating sparks and starting fires. F-202604-066061

Timeline	Remediation Plan
Medium Term CAP-0078768	Engage a qualified lightning protection specialist to perform a comprehensive risk assessment as per BS EN 62305. This assessment will determine the necessity of a lightning protection system (LPS) based on factors such as building height, location, occupancy, and the presence of sensitive electronic equipment.
Longer Term CAP-0078769	If the risk assessment concludes that an LPS is necessary, proceed with the design and installation of an appropriate system in compliance with relevant Standards such as BS EN 62305 (Protection against Lightning).

8. Fire Alarm & Detection System (FADS)

8.1 Non-functional or incomplete

This section refers to the entire facility.

During the inspection, the Fire Alarm and Detection System (FADS) for the facility was evaluated.

This section refers to the entire facility.

An addressable fire alarm system was installed in the facility in the security room that serves the apparel division of the facility. During the inspection, several deficiencies were noted, including the use of non-listed panel and components, inadequate detection coverage, missing manual call points (MCP's), fault indication on panel, faulty manual call points (MCP's), 220V manual hooters, non-fire rated cabling, incorrect placement of manual call points (MCP's) and detection devices, low audibility in several areas during testing, all of which pose a significant life-safety risk during a fire emergency.

A non-functional/incomplete FADS significantly increases the risk of worker injury or fatality, substantial property damage, operational disruptions, legal liabilities, reputational damage, and environmental harm. A delayed response during a fire event can significantly increase the impact on personnel, goods, and property.F-202604-066059

Timeline	Remediation Plan
Medium Term CAP-0078976	i. Submit design documentation for the new replacement of FADS system to Pakistan Accord for review before purchasing any new equipment. Design documentation to include drawings, technical data sheets, and product certifications of all listed equipment to show full compliance with the Fire Alarm and Detection System (FADS) with NFPA 72.
Longer Term CAP-0078977	ii. Submit documentation to the Pakistan Accord showing that an L/C has been opened for the import of any components required for the FADS design submitted earlier.
Longer Term CAP-0078978	iii. Submit documentation to the Pakistan Accord demonstrating that all materials, both imported and locally procured, have been purchased and are available on-site for installation.

Longer Term CAP-0078979	iv. Once installation of the FADS system is complete, the factory is to submit full "as-built" documentation of the system to the Pakistan Accord for review.
Longer Term CAP-0078980	v. Factory to advise Pakistan Accord when FADS installation is fully completed and ready for review. Pakistan Accord will then arrange a verification inspection to ensure that the installation fully complies with the previously reviewed design documentation.
Longer Term CAP-0078981	vi. Provide full testing and commissioning (T&C) certification from the FADS system installer and make provision for attendance of the system installer, factory technical personnel, and Pakistan Accord technical team to witness a final T&C demonstration verification on site. On successful completion of this T&C verification inspection, remediation of the FADS system will be complete.



Figure 2.36 - Addressable fire alarm panel at security office with fault indication



Figure 2.37 - Manual call points (MCP's) missing at entrance of main building



Figure 2.38 - Incorrectly installed manual call point (MCP) at cutting hall main building



Figure 2.39 - Incorrectly installed smoke detectors at stitching area main building



Figure 2.40 - No detection coverage in compressor room



Figure 2.41 - No detection coverage in maintenance workshop main building

9. Emergency Lighting

9.1 Provided but non-functional

The section refers to the entire facility.

During the inspection it was observed that emergency lighting was provided but not functional in a number of locations such as packing area ground floor, finishing area ground floor, stitching area first floor and sample room first floor main building.

Functional emergency lighting is critical for providing illumination in the event of a power failure or other emergencies, ensuring that workers can safely evacuate the facility.F-202604-066067

Timeline	Remediation Plan
Medium Term CAP-0078778	Conduct a thorough inspection of all emergency lighting units throughout the facility to identify specific locations and units that are non-functional. This involves checking both the lights and the power sources, including batteries and wiring connections. Repair any faulty emergency lighting units immediately. If repair is not feasible, replace the defective units with new, compliant emergency lighting fixtures to ensure they provide adequate illumination during an emergency.
Longer Term CAP-0078779	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.42 - Non functional emergency light at packing area ground floor main building



Figure 2.43 - Non functional emergency signage light at finishing area ground floor main building



Figure 2.44 - Non functional emergency light at stitching area first floor main building



Figure 2.45 - Non functional emergency light at sample room first floor main building

9.2 Provided but insufficient

The section refers to the entire facility.

During the inspection the emergency lighting arrangement throughout the facility was assessed.

The following observations were made:

i. Several areas lack sufficient emergency and directional exit lighting coverage including but not limited to the following areas:
a) Compressor room
b) Cutting hall
c) Pressing and packing area
d) Stairs area
e) Stitching area
ii. External emergency lighting was not provided, preventing people from reaching a place of safety outside the facility during nighttime evacuations.

Adequate and functional emergency lighting throughout the facility is crucial for providing necessary illumination during power failures or other emergencies, ensuring the safe evacuation of workers.F-202604-066060

Timeline	Remediation Plan
Medium Term CAP-0078780	i. Conduct a comprehensive survey to identify all areas lacking adequate emergency lighting and exit signages; develop design layout drawings in accordance with the Building Code of Pakistan Fire Safety Provisions 2016, submit them to Pakistan Accord for review, and install additional emergency luminaires and illuminated exit signages as per the approved layout.
Medium Term CAP-0078781	ii. Install external emergency luminaires to enable safe evacuation during hours of darkness all the way to a safe place outside the building(s)

Longer Term CAP-0078782	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.46 - Insufficient emergency and directional exit lighting at stitching area main building



Figure 2.48 - No external emergency lighting in assembly area



Figure 2.47 - No emergency lighting in stairs area main building



Figure 2.49 - Insufficient emergency and directional exit lighting at cutting hall main building

9.3 Independent power source

The section refers to the entire facility.

During the inspection, it was observed that the emergency lighting within the facility was not connected in a manner that would ensure automatic activation upon the loss of power to the general lighting in various spaces.

Proper circuitry is crucial to ensure that emergency lighting is automatically activated when general lighting fails. This automatic activation is essential for maintaining sufficient illumination, allowing for the safe evacuation of occupants and preventing panic during power outages. Further the emergency lights and exit signs should be reconfigured to be connected on a dedicated single circuit, which must not include any other load except for the emergency lighting system. According to the Building Code of Pakistan Fire Safety Provisions 2016, emergency lighting systems must be designed to provide immediate illumination in the event of power loss to ensure a clear and safe path for evacuation.

Ensuring that the emergency lighting is integrated with the facility's power system to activate automatically during power failures is critical for life safety. This integration helps prevent hazardous conditions that can arise in the dark, such as trips, falls, and confusion, thereby facilitating orderly and efficient evacuation during emergencies.F-202604-066064

Timeline	Remediation Plan
Medium Term CAP-0078791	Reconfigure the emergency lighting circuitry to ensure that emergency lights activate immediately when general lighting power fails, as required by the Building Code of Pakistan Fire Safety Provisions 2016. Emergency lighting shall be powered from a source independent of that supplying the normal lighting.
Longer Term CAP-0078792	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.

10. Cabling & Wiring

10.1 Cable trench and channel cleanliness

This related to the whole facility.

During the inspection of the facility, it was observed that the cleanliness of the cable trunking system had not been maintained. Numerous trunking systems were found to be filled with a significant amount of fluff and dust, which poses a serious fire risk.

This issue was observed near main distribution board ground floor and outside main building.

Accumulated dust and fluff can ignite when exposed to sparks or high temperatures, potentially leading to a fire hazard. Maintaining a clean cable trunking system is essential to minimise fire risks associated with accumulated debris.F-202604-066054

Timeline	Remediation Plan
Medium Term CAP-0078885	Clean all cable trunking systems to remove accumulated dust and fluff, ensuring that any potential fire hazards are mitigated.
Longer Term CAP-0078886	Implement a regular maintenance schedule that includes routine inspections and cleaning of the cable trunking systems to prevent future accumulation of dust and fluff, thereby maintaining a safe environment and complying with safety standards.



Figure 2.50 - Cable trench filled with dust and debris at ground floor main building



Figure 2.51 - Cable trench filled with dust and debris outside main building

10.2 Splicing of cables

This relates to the whole facility.

During the inspection the cable installation was visually surveyed throughout the facility.

The following observations were noted:

i. Cables terminations within panels and switchgears were made without cable lugs, resulting in breaking strands and loose connections. This presents a significant fire risk due to the potential for electrical arcing and overheating, which can ignite nearby materials and cause a fire. Furthermore, loose connections can cause voltage drops and equipment malfunctions, posing a life safety hazard. This issue was observed at finishing area ground floor, main distribution boards, compressor room, cutting hall ground floor and stitching area first floor of main building.

ii. PVC tape joints were found in several locations - including within distribution boards - where cables were either damaged or two cables were being joined together. This method of jointing is unreliable and can deteriorate over time, increasing the risk of electrical shorts and fires. The adhesive on PVC tape can degrade, leading to exposed conductors that can spark and ignite surrounding materials, presenting both fire and life safety risks. This issue was specifically observed at compressor room, cutting hall ground floor main building and first floor stitching area main building.

iii. Low quality PVC connectors were used in distribution boards which could be hazard of potential short circuit due to its poor insulation. This issue was specifically observed at cutting hall ground floor main building.

iv. Open, exposed, and idle cable connections were found inside distribution boards, posing a risk of electric shock, short circuit, and fire. This issue was specifically observed at first floor stitching area main building.

For any future electrical work, the termination of stranded cables must be done with cable lugs and PVC tape joints shall not be utilized. Proper termination ensures secure and reliable connections, reducing the risk of electrical failures and enhancing overall safety.F-202604-066053

Timeline	Remediation Plan
Medium Term CAP-0078900	i. Install cable lugs on all improperly terminated cables within panels and switchgears to secure connections and prevent breaking strands and loose connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022)

Longer Term CAP-0078907	i. Implement a standard operating procedure (SOP) requiring the use of cable lugs for all future cable terminations. Regularly train and audit personnel to ensure compliance with this procedure, thus maintaining safe and reliable electrical connections.
Medium Term CAP-0078908	ii. Replace all PVC tape joints with proper terminal blocks or connectors to ensure secure and stable connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022). Replace all damaged cabling with new wiring.
Longer Term CAP-0078909	ii. Establish a policy prohibiting the use of PVC tape for electrical jointing. Regularly inspect and maintain electrical installations to ensure compliance, and provide continuous training for staff on proper jointing techniques.
Medium Term CAP-0078910	iii. Replace the PVC connector with a standard quality terminal block in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022).
Longer Term CAP-0078911	iii. Establish a plan to replace all PVC connectors with compliant electrical terminal blocks and integrate this into regular maintenance to ensure ongoing safety and compliance.
Medium Term CAP-0078912	iv. Remove unused cables or properly terminate and insulate open and exposed connections to eliminate the risk of sparks and potential fires. (Reference: BS 7671 18th Edition + A2 (2022) Regulation 416.1 for basic insulation of live parts).
Longer Term CAP-0078913	iv. Establish routine inspections to ensure all electrical terminations remain properly insulated and protected from exposure, especially in high-risk operational areas.



Figure 2.52 - Cable termination without lugs at finishing hall ground floor main building



Figure 2.53 - Idle and open cables found at first floor stitching area main building



Figure 2.54 - PVC connector, tape joints in panel at cutting hall main building



Figure 2.55 - PVC tape joints observed at compressor room panel

10.3 Colour coding

This relates to the whole facility.

During the inspection of several distribution boards, deviations from standard cable colour coding practices were identified. Such deviations pose significant risks, as they can lead to the misinterpretation and misidentification of cables. This misidentification can result in incorrect connections, potentially causing unsafe working conditions and increasing the likelihood of electrical accidents, such as shocks or fires.

This issue was observed at compressor room, cutting hall main building, canteen building and fire pump area panel.

To mitigate these risks, it is imperative that all future electrical work strictly adheres to standardized colour coding as prescribed by BS 7671. This will ensure accurate identification of conductors, thereby enhancing overall safety and operational efficiency. F-202604-066048

Timeline	Remediation Plan
Medium Term CAP-0078936	Conduct a comprehensive audit of all distribution boards to identify and correct any deviations from the standard cable colour coding. This involves re-labelling or re-insulating the cables with the appropriate colour codes as specified in BS 7671. Specifically, adhere to BS 7671 Regulation 514.3.1, which mandates the correct identification of conductors.
Longer Term CAP-0078937	Develop and enforce a strict protocol for cable colour coding to be followed for all future installations and maintenance activities. This protocol should include regular training sessions for electricians and maintenance personnel to keep them updated on the latest standards and practices. Additionally, periodic audits should be conducted to ensure ongoing compliance with the standardized colour coding system, thereby preventing future deviations and ensuring sustained safety and operational efficiency.



Figure 2.56 - Colour coding violation in distribution board at compressor room



Figure 2.57 - Colour coding violation in distribution board at canteen building



Figure 2.58 - Colour coding violation in distribution board at fire pump area

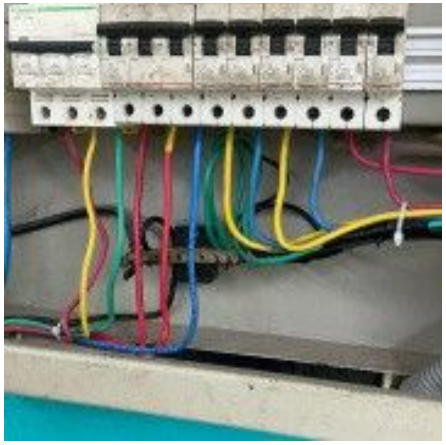


Figure 2.59 - Colour coding violation in distribution board at cutting hall main building

10.4 Fixtures

This relates to the whole facility.

During the inspection, the electrical fixtures throughout the facility were visually surveyed.

The following observations were noted:

i. Power receptacles mounted on combustible wooden ply: Installing receptacles on combustible materials like wooden ply increases the risk of fire if a short circuit or overheating occurs. Combustible materials can easily ignite, leading to potential fire hazards and endangering life safety. This issue was systematic in most areas of the facility.

ii. LED lights were found suspended directly from cables, and several fixtures lacks proper support, posing a risk of cable strain, electrical faults, or fire hazards. This issue was observed at cutting hall main building and maintenance workshop. F-202604-066056

Timeline	Remediation Plan
Medium Term CAP-0078951	i. Replace combustible wooden ply backings with non-combustible materials to reduce fire risk.
Longer Term CAP-0078952	i. Establish a facility-wide standard to use only non-combustible materials for mounting electrical components, and conduct regular audits for compliance.
Medium Term CAP-0078953	ii. Install appropriate lighting fixtures with secure mounting and replace unsafe wire supports with proper systems to ensure safe and compliant installation.
Longer Term CAP-0078954	ii. Standardize facility-wide safe lighting installation using durable supports, conduits, and proper cable management to prevent wiring stress and ensure safety.



Figure 2.60 - Light fixture on cable weight at maintenance workshop main building



Figure 2.61 - Power receptacles on wooden board at compressor room



Figure 2.62 - Light fixture without proper support at cutting hall main building

11. Assessment Limitations and Assumptions

The information in this report was obtained during an inspection of the facility and based on information provided by factory management. It has not been possible to provide independent verification for all the information and data collected, and therefore, the Accord cannot accept general responsibility for omissions or errors arising from inaccuracies in this report from the information obtained.

The findings and recommendations in this report are not intended to imply, guarantee, ensure or warrant compliance with any government regulations. Additionally, the results do not imply in any way that compliance with the findings or recommendations as stated in this report will eliminate all hazards, risks or exposures or that hazards, risks or exposures not referred to in this report do not exist. Compliance with the findings and recommendations stated in this report does not relieve the factory owner from obligation to comply with specific project requirements, industry standards, or the provisions of any local government regulations.

Except as otherwise noted within this report, other systems or services were not viewed or tested during the assessment and are therefore similarly excluded from this report. We have not assessed any parts of the structure / building fabric / services which are covered, unexposed or inaccessible and we are therefore unable to report that any such part of the property is free from defect.

The observations in this report are based on the engineering judgement of the engineer at the time of the inspection.